

Why I selected this algorithm

The Caesar cipher was selected from the *Cryptography with Python* Tutorialspoint resource due to its conceptual simplicity and suitability for demonstrating fundamental programming and cryptographic principles. The algorithm applies a fixed positional shift to alphabetic characters while preserving non-alphabetic input, making it both predictable and easy to validate during development (Tutorialspoint, no date). This clarity allowed the focus to remain on core programming constructs such as string manipulation, control flow, and file input/output, rather than the complexities associated with external cryptographic libraries.

From a design perspective, the implementation was intentionally kept lightweight and self-contained. The program accepts user input for short text encryption and extends this functionality to process entire text files, producing an encrypted output file. This directly satisfies the seminar requirement while ensuring the code remains transparent enough to explain step-by-step in a live demonstration.

However, this choice also reflects a deliberate trade-off. While the Caesar cipher is effective as a teaching tool, it lacks the complexity required for real-world security. Therefore, its inclusion here is best understood as an educational prototype that supports understanding of transformation-based encryption, rather than as a secure solution suitable for practical deployment.

Would it meet GDPR regulations?

This implementation would not meet the requirements of the General Data Protection Regulation (GDPR) for protecting personal data. Article 32 states that organisations must implement “appropriate technical and organisational measures” to ensure a level of security appropriate to the risk, explicitly including encryption as a potential control (European Parliament and Council, 2016). However, the effectiveness of such measures depends on their resistance to realistic attack scenarios.

The Caesar cipher provides only a minimal level of protection, as it relies on a limited key space of 25 possible shifts. This makes it trivial to break using brute-force methods or frequency analysis, meaning it does not offer meaningful confidentiality for modern data. As such, it cannot be considered an appropriate encryption method under GDPR.

Furthermore, compliance extends beyond algorithm selection. Effective data protection also requires secure key management, access controls, governance policies, and ongoing risk assessment. While this prototype demonstrates the principle of data transformation and supports the concept of data protection by design, it would only contribute to GDPR compliance as part of a broader, well-implemented security framework. On its own, it is insufficient.

Reference list (Harvard style)

European Parliament and Council (2016) *Regulation (EU) 2016/679 (General Data Protection Regulation)*. Official Journal of the European Union, L119, pp. 1–88. Available at: <https://www.wipo.int/wipolex/en/text/480507>.

Tutorialspoint (no date) *Cryptography with Python – Caesar Cipher*. Available at: https://www.tutorialspoint.com/cryptography_with_python/cryptography_with_python_caesar_cipher.htm.